P.S.R. ENGINEERING COLLEGE, SIVAKASI – 626 140
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
QUESTION BANK

Programme:	B.E.	Branch	Electronics and Communication Engineering
Academic Year:	2025-2026	Year/Semester	II/IV
Course Code:	231EC45	Course Name	JAVA Programming
Course Tutor:	Mr.D.Arun Shunmugam, AP/ECE Ms.A.Madhumathi AP/ECE		

UNIT IV: GRAPHICAL USER INTERFACE (GUI)

PART-A			
S.NO	**QUESTION**	**BTL**	**CO**
	Which of the following is a feature of the AWT (Abstract Window Toolkit) in Java? A) Lightweight components B) Heavyweight components C) No event handling D) Cannot be used for GUI development Answer: b	K1	CO4
	Which of the following layout managers arranges components in a grid of rows and columns? A) FlowLayout B) BorderLayout C) GridLayout Answer: c	K1	CO4
	Which method is called when an applet is first loaded into memory? A) start() B) init() C) stop() D) destroy() Answer: b	K1	CO4
	Which of the following is true about Swing components? A) Swing components are heavyweight. B) Swing components are dependent on the underlying operating system. C) Swing components are written entirely in Java. D) Swing components are only used for event handling. Answer: c	K1	CO4
	What is the purpose of the ActionListener interface in Java? A) To handle mouse events. B) To listen for key events. C) To handle button clicks and other action events. D) To manage window resize events. Answer: c	K1	CO4
	In an applet, how do you pass parameters through an HTML tag? A) Using <param> tag within <applet> tag. B) Using the setParameter method. C) Parameters are passed automatically. D) By defining variables within the applet's Java code Answer: a	K1	CO4
	Which method of the JFrame class is used to set the window's title? A)setTitle() B) setWindowTitle() C) windowTitle() D) title() Answer: a	K1	CO4
	Which of the following methods in an applet is responsible for cleaning up resources when the applet is destroyed? A) start() B) init() C) stop() D) destroy() Answer: d	K1	CO4

	Which of the following components is a container in Swing? A) JLabel B) JButton C) JTextField D) JPanel Answer: d	K1	CO4
	Which of the following Swing components is used to create a button? A) JButton B) JTextField C) JList D) JCheckBox Answer: a	K1	CO4
	Which layout manager arranges components in five regions: North, South, East, West, and Center? A) GridLayout B) BorderLayout C) FlowLayout D) CardLayout Answer: b	K1	CO4
	Which method is used to make a JFrame visible? A) show() B) visible() C) setVisible(true) D) display() Answer: c	K1	CO4
	What is the default layout manager for a JPanel in Swing? A) BorderLayout B) FlowLayout C) GridLayout D) CardLayout Answer: b	K1	CO4
	Which of the following is not a method of the ActionListener interface? A) actionPerformed() B) actionOccurred() C) addActionListener() D) removeActionListener() Answer: b	K1	CO4
	Which component is used to create a drop-down list in Swing? A) JList B) JComboBox C) JRadioButton D) JCheckBox Answer: b	K1	CO4
	Which of the following is a valid layout manager in AWT? A) FlowLayout B) GridBagLayout C) BorderLayout D) All of the above Answer: d	K1	CO4
	What is the purpose of the ActionListener interface in Java? A) It handles keyboard events. B) It handles mouse events. C) It handles action events such as button clicks. D) It handles window events Answer: c	K1	CO4
	Which method is automatically called when an applet is started? A) init() B) start() C) stop() D) destroy() Answer: b	K1	CO4
	Which of the following is a container class in Swing? A) JButton B) JTextField C) JPanel D) JLabel Answer: c	K1	CO4
	Which method in AWT is used to handle button clicks? A) buttonPressed() B) actionPerformed() C) handleClick() D) actionEvent() Answer: b	K1	CO4

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PART B(2-Mark Questions)			
S.NO	**QUESTION**	**CO**	**BTL**
1.	What is AWT in Java, and how does it differ from Swing?	CO4	K1
2.	List the basic components provided by AWT for GUI development.	CO4	K1
3.	Explain the difference between lightweight and heavyweight components in Java.	CO4	K1
4.	Describe the role of a layout manager in AWT and Swing.	CO4	K1
5.	Write a simple Java program using AWT to create a window with a button.	CO4	K1

6.	How do you handle button click events using ActionListener in AWT?	CO4	K1
7.	Compare and contrast the FlowLayout and BorderLayout layout managers in AWT.	CO4	K2
8.	Analyze the life-cycle of an applet and explain each of its stages.	CO4	K2
9.	Evaluate the performance differences between AWT and Swing components for GUI applications in Java.	CO4	K2
10.	Assess the role of JApplet in modern web development and its alternatives.	CO4	K2
11.	What is the purpose of the init() method in the applet life-cycle?	CO4	K1
12.	What is the difference between the JButton and JCheckBox in Swing?	CO4	K1
13.	Explain how the JComboBox component is used to create a drop-down list in Swing.	CO4	K2
14.	How do you pass parameters from an HTML page to an applet using the <param> tag?	CO4	K1
15.	Write an AWT program to demonstrate the use of multiple layout managers in a single window.	CO4	K1
16.	Demonstrate how you can embed an applet in an HTML page with parameters.	CO4	K2
17.	Compare between JPanel and JFrame in terms of functionality in a Swing-based application.	CO4	K2
18.	How would you implement event handling for a JTextField to perform an action when the user presses the "Enter" key?	CO4	K1
19.	Create a Java Swing application that uses a JFrame to display a JButton and JLabel. When the button is clicked, the label’s text should change.	CO4	K2
20.	Design a Swing form using JTextField, JButton, and JLabel to accept a user's name and display it upon clicking a button.	CO4	K2

PART C 14 Mark Questions			
S.No	**QUESTION**	**CO**	**BTL**
1.	Explain the differences between **AWT** and **Swing** in Java.(7) **b)** Describe the types of components available in **AWT** and give examples of when each might be used.(7)	CO4	K3
2.	**a)** Develop a Java program using **AWT** to create a simple user interface that contains a **text field** and a **button**. When the button is clicked, the text entered in the text field should be displayed in a **label** below the text field.(7) **b)** Analyze the event handling process in the above example. How is the button click event captured and processed?(7)	CO4	K2
3.	**a)** Design a Java program that uses **Swing components** to create a login form with two JTextFields (for username and password) and a JButton (for login). When the button is clicked, the entered username and password should be displayed in a dialog box.(7) **b)** Explain how the **Swing layout manager** is used to position the components within the JFrame.(7)	CO4	K3
4.	**a)** Describe the **Applet life-cycle** with respect to the following methods: init(), start(), stop(), and destroy().(7) **b)** Write a small program using **Applet** that demonstrates the use of these methods and shows how parameters can be passed from an HTML page to the applet.(7)	CO4	K3
5.	**a)** Discuss the advantages and disadvantages of using **AWT** vs **Swing** for building modern Java GUI applications.(7) **b)** Design a GUI using **Swing** to demonstrate the use of **JComboBox**, **JRadioButton**, and **JCheckBox**. The GUI should allow the user to select a color from the JComboBox and use checkboxes to select options. Display the selections when the button is clicked.(7)	CO4	K3
6.	Explain the concept of **layout managers** in Java. Compare the **FlowLayout**, **BorderLayout**, and **GridLayout** managers in terms of their use cases and how they differ in arranging components.(7) **b)** Create a user interface using **Swing** that demonstrates the use of **BorderLayout** and **GridLayout** together to arrange buttons and text fields in a panel.(7)	CO4	K2
7.	**a)** Discuss the **Swing** component JFrame in terms of creating a window-based application. What are its key features, and how does it interact with other Swing components?(7) **b)** Create a simple **Swing-based application** that uses **JFrame** to display a title, a text input field, and a button. When the button is clicked, the text entered in the field should be displayed in a message dialog.(7)	CO4	K3